AUTHOR TESTED VIDEO PROOF PROVIDED

Simple Stereo FM RADIO RECEIVER

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There are many AM/FM receiver circuits available, but the difficult part is making the coils and tuning the circuits. Without their proper construction and alignment, the circuits do not perform well. This article describes how to build a simple stereo FM receiver without any tuning elements such as inductors, variable capacitors, and ceramic IF filters.

The receiver is capable of receiving 76 to 108MHz FM band with stereo output having good sensitivity and performance. The output is capable of driving a 32Ω speaker directly, which eliminates additional audio headphone amplifier.

The heart of the circuit is based on IC HEX3653/GS1299/RDA7088. We can use any one of them. All three ICs have the same pinout and function and come in SOP16 package with 1.27mm lead pitch.

BILL OF MATERIAL

Ref.	Descriptions of the components	Qty
U1	RDA7088, HEX3653, or GS1299 in SOP16 package	1
Y1	32.768kHz crystal (2mm x 6mm or 3mm x 8mm)	1
R1	Resistor 10kΩ , 250mW (CFR 5%)	1
C3, C5	Capacitor 100nF	2
Antenna	75cm SWG#22	1
C1, C2	Capacitor electrolytic, 100µF/16V	2
L1	Inductor 100nH	1
C6	Capacitor 24pF	1
	Breadboard or Veroboard	1
	SOP16 1.27mm to 2.54mm adaptor, or ZIF socket	1
	Jumper wires	
	3V DC or AMS1117-3.3	

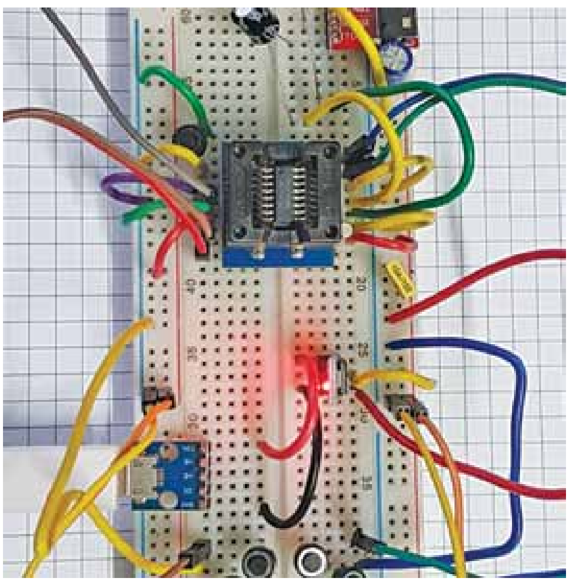

Fig. 1: Author's prototype

We can either use a pitch converter breakout board for ease of using a breadboard, Veroboard, or dotboard with 2.54mm pitch. We can also use SOIC-16 ZIF socket adaptor with 2.54mm pitch for easy and quick experimentation.

Fig. 1 shows the author's prototype. The components required to build the radio circuit are mentioned under the bill of material table.

As can be seen, the radio receiver requires very few external components, yet supports the worldwide frequency band of 76 to 108MHz. Its other features include:

- Less than 5-second full band seek time
- Low-cost 32.768kHz crystal for RCLK
- Digital automatic gain control (AGC)
- Direct 32Ω speaker driving
- Simple SOP16 package
- 1.8V to 3.6V DC voltage operation
- Low current consumption of 17mA at 3V

Circuit diagram of the FM receiver circuit is shown in Fig. 2. The receiver is mainly built around the FM RDA7088 IC along with a few passive components.

Construction and testing

Assemble the circuit on Veroboard or breadboard or a general-purpose PCB. Refer Fig. 1 through Fig. 4 before assembly. First, fix the IC using a 2.54mm adaptor board or ZIF SOIC-16 socket. An adaptor board requires skill to solder the 16-pin, 1.27mm pitch IC.

Connect all the GND (2, 3, 5, 11, and 14) and VDD (6 and 10) pins, as shown in the diagram. Connect the 32.768kHz crystal's one end to pin 9 and the other to GND.

Connect push-button switches (5 in all) to pin numbers 1, 7, 8, 15, and 16 for POWER ON/OFF, SEEK-, SEEK+, VOL-, and VOL+ functions, respectively. Resistor R1 (10kΩ) and capacitor C3 (100nF) are optional and not necessary.

Connect one end of capacitors C1 and C2 to pin 12 and pin 13 and their other end directly to 32Ω speaker's + terminal or to a 3.5mm audio jack for connecting an earphone or headphone of 32Ω resistance. Connect capacitor C5 to pin 10 and GND.

For antenna, connect 75cm length of SWG#22 wire to pin 4 of the IC. Inductor L1 and capacitor C6 form the band rejector as their parallel combination allows 76 to 108MHz band and rejects frequencies below 76MHz and above 108MHz. L1 and C6 improve performance but are not mandatory if availability is an issue.